

NLP In Deep Learning

① ANN → ARTIFICIAL NEURAL N/W → Tabular Data.

House size No. of Rooms Price



↑ Order is Important
Sequential Data
vs
Non Sequential Data

② CNN → Images → Image classification, Object Detection

⇒ Sequence of Data IS Important

③ i) RNN

ii) LSTM RNN

iii) GRU RNN

iv) Encoder Decoder

v) Attention is all you need

⇒ Sequential Data. [NLP] [Time Series]

Eg: Chat bot App → Q & A

Language Translation → Eng → French

Text Generation → A sentence → Completion of sentence

Auto Suggestion → LinkedIn, Gmail

Time Series → Sales Data Future Prediction.

① Can we solve with ANN?

Eg:

<u>Text</u>	<u>o/p</u>
The <u>food</u> is good	1
The food is <u>bad</u>	0
The food is <u>not</u> good	0

① Words → Vectors
↓ Sequential Info.

Bow, TF-IDF [Context Is Missing].

	Word 1	Word 2	Word 3	-	-
Sent 1	0	1	0	0	0
Sent 2	1	1	0	0	0
Sent 3	0	0	1	0	0

RNN

① Timestamps

② Inp Text

Text

o/p

The food is good

1

The food is bad

0

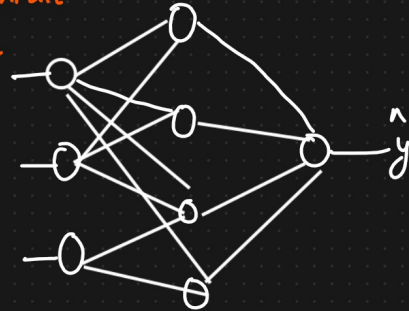
The food is not good

0

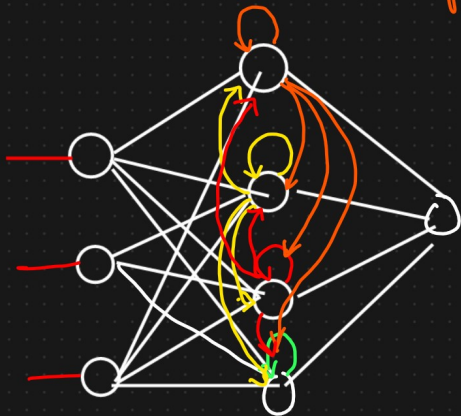
RNN Architecture



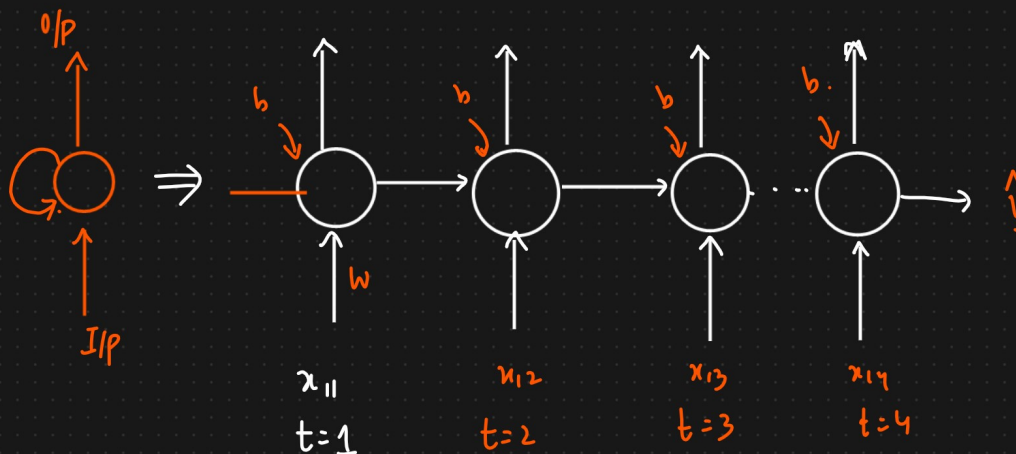
I/p is sent all at once



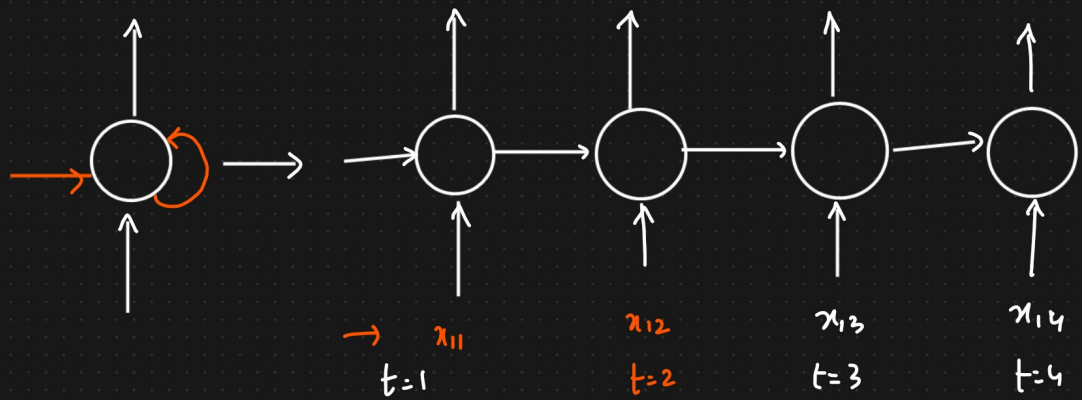
RNN



x_{11} x_{12} x_{13} x_{14}
→ The food is good



Sentence words



Working of Simple RNN With Forward Propagation

Dataset

Text

$x_{1,1}$ $x_{1,2}$ $x_{1,3}$
The food is good
The food is bad
The food is not good

O/P

1

0

0

Words $\rightarrow$ Vectors

OHE $\rightarrow$ One hot Encoding

$\begin{bmatrix} [1 & 0 & 0 & 0 & 0] & [0 & 1 & 0 & 0 & 0] & [0 & 0 & 1 & 0 & 0] \\ [0 & 0 & 0 & 1 & 0] & [0 & 0 & 0 & 0 & 1] \end{bmatrix}$



$t=1$ $x_{1,1}$

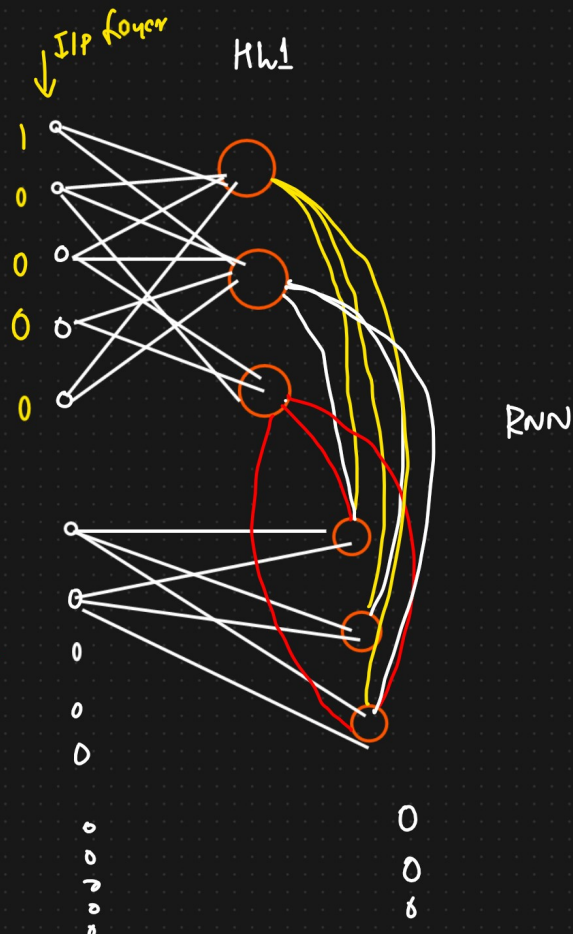
$=$

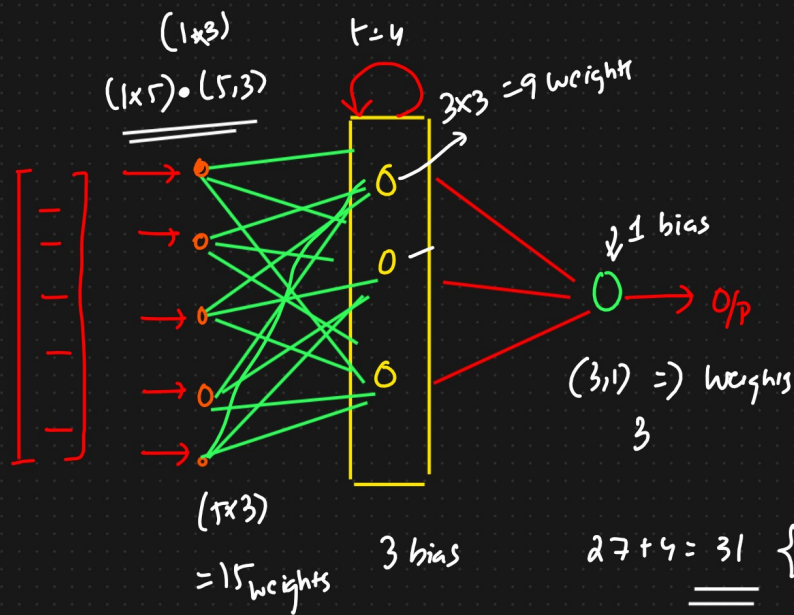
The

$t=2$ $x_{1,2}$

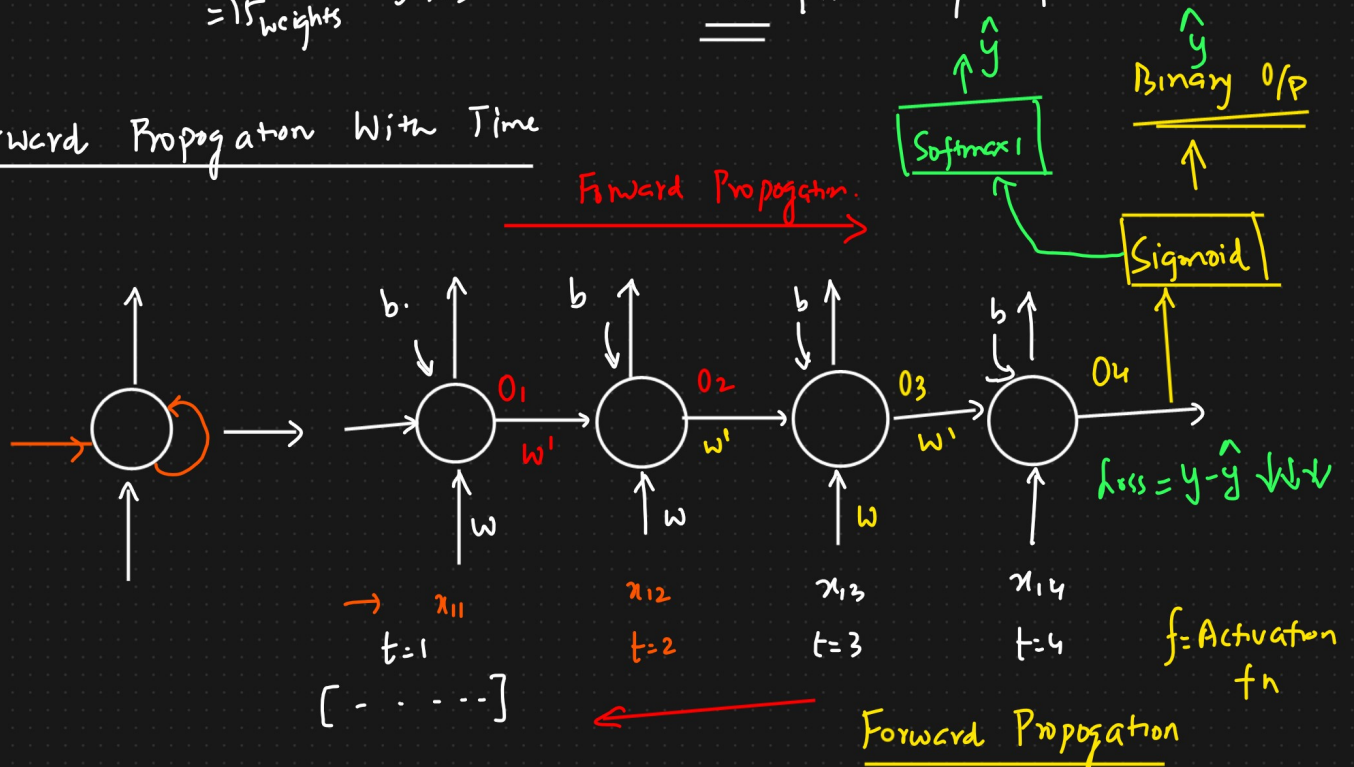
$=$ food

$t=3$





Forward Propagation With Time



Data at

The food is good
 $x_{11} \quad x_{12} \quad x_{13} \quad x_{14}$

O/P
 1

$$O_1 = f(x_{11} \cdot w + b_1)$$

$$O_2 = f(x_{12} \cdot w + O_1 \cdot w' + b_1)$$

$$O_3 = f(x_{13} \cdot w + O_2 \cdot w' + b_1)$$